

Probability Distributions

Lecture/Lab 04

Physics 4730, Fall 2026



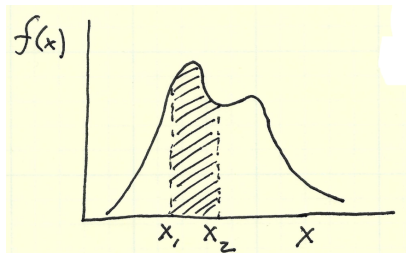
IIDRV's

- ▶ **Random Variable (RV)** – value results from measurements of a quantity subject to random variations.
- ▶ RV types:
 - ▶ Discrete
 - ▶ Continuous
- ▶ **Independent and Identically Distributed (IID)** – variables drawn from the same distribution, that are independent.

→ IIDRV if $p(x_1, x_2) = p(x_1)p(x_2)$

Probability and Probability Density

In a continuous distribution, the probability of measuring any *particular* value is zero! The best we can do is define a “probability density function” (PDF) $f(x)$ such that:



$$\int_{x_1}^{x_2} f(x) dx = P(x_1 < x < x_2)$$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Descriptive Statistics (location, scale, shape of PDF)

Mean (expectation value):

$$\mu = E = \int_{-\infty}^{\infty} x f(x) dx$$

Variance (standard deviation)²:

$$\sigma^2 = V = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

Skewness (asymmetry):

$$\Sigma = E = \int_{-\infty}^{\infty} \left(\frac{x - \mu}{\sigma} \right)^3 f(x) dx$$

Kurtosis (heavy-tailedness):

$$K = \int_{-\infty}^{\infty} \left(\frac{x - \mu}{\sigma} \right)^4 f(x) dx - 3$$

→ “moments” of distribution

(kurtosis of a normal distribution $\equiv 0$)

A couple of other descriptors...

Absolute deviation about d :

$$\delta = \int_{-\infty}^{\infty} |x - \mu| f(x) dx$$

Mode (most probable value):

$$\left. \frac{df(x)}{dx} \right|_{x=x_m} = 0$$

Estimates of Descriptors from N Data Points

Mean $\mu = \int_{-\infty}^{\infty} x f(x) dx \rightarrow \bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$

StdDev $\sigma = \sqrt{\int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx} \rightarrow S = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$

Mean Err $\sigma_{\bar{x}} = \frac{S}{\sqrt{N}}$

StdDev Err $\sigma_S = \frac{S}{\sqrt{2(N-1)}}$

Lab 4: Exponential Distribution

PHYSICS 4730
Lab 04

- [Exercises](#)
- [Assignment](#)

Exercise 1.

Please review the [random variable](#) notebook that the instructor introduced in lecture.

Exercise 2.

Prepare a standalone Python script **expo.py** to perform the following:

- Read data from [expodata.dat](#), assumed to be in the current working directory.
- Have your code determine and write to standard output the parameter β , as inferred from the *mean* and *variance* of the distribution. (You'll have to look up or calculate the mean and variance of an exponential distribution.)
- As a comment in your code, answer the following question: Is it possible to determine the parameter β from the skewness or kurtosis of the distribution? Why or why not?
- Plot the data as a histogram showing the *empirical probability density function*. Superimpose on this the pdf according to `scipy.stats`, assuming the value for β that you determined above.

Your code should run on the Physics Department Linux Cluster, with the command **python3 expo.py**.

Assignment, due 3 September 2026 in class

Problem 1

Submit your code **expo.py** from Exercise 2 above.

`submit p4730 lab04 expo.py`

note: Python convention

$$f(x) = \frac{1}{\beta} e^{-x/\beta}$$

versus

$$f(x) = \beta e^{-\beta x}$$

check normalization

$$\int_0^{\infty} \frac{1}{\beta} e^{-x/\beta} dx = \frac{1}{\beta} \left[-\beta e^{-x/\beta} \right]_0^{\infty} = 0$$

rv_gallery.ipynb